Periodic solutions for nonlinear differential equations in Banach spaces with generalized variable-exponent operators

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Motivation and the main problem

Periodic models in ordinary differential equations

Many systems (physical, biological, among others) are subject to recurrent effects, such as periodic forcing, cyclic loading, or oscillatory behaviours.

Usually, a simple way of modeling these phenomena is by second-order ordinary differential systems

$$u''(t) = f\big(t, u(t), u'(t)\big), \qquad u : [0, T] \to \mathbb{R}^N.$$

For these models, we seek solutions satisfying

$$u(T) = u(0), \qquad u'(T) = u'(0),$$

so that the system returns to the same state and velocity after one period.

Unlike IVP, the initial state is not prescribed: it must be chosen so that the trajectory closes after one period. Thus, periodic problems are difficult and have been a classical motivation for several methods of nonlinear analysis, including topological degree and continuation techniques.

From classical equations to nonlinear constitutive laws

The previous model assumes a linear response with respect to the velocity, since the principal term is u''.

In nonlinear models, this response may instead be described by a constitutive law ϕ, leading to

$$\frac{d}{dt}\phi\big(u'(t)\big) = f\big(t, u(t), u'(t)\big).$$

Such problems arise, for instance, in non-Newtonian fluids and nonlinear elasticity, among others.

More generally, the constitutive response may also vary with time. For instance, a time-dependent coefficient may lead to

$$\frac{d}{dt}\Big[a(t)\phi\big(u'(t)\big)\Big] = f\big(t, u(t), u'(t)\big),$$

Previous theory in finite dimensions

Both time-independent and time-dependent constitutive laws can be conveniently included in periodic problems of the form

$$\begin{cases} \dfrac{d}{dt}S(t,u'(t)) = f(t,u(t),u'(t)), \\ u(0) = u(T), \qquad u'(0) = u'(T), \end{cases} \qquad u : [0,T] \to \mathbb{R}^N.$$

Problems driven by nonlinear operators of this type have been studied in several different settings, including recent works on generalized variable-exponent operators [8, 9, 13].

This naturally raises the question of extending this theory to infinite-dimensional state spaces.

From $\mathbb{R}^N$ to a Banach space

Such systems arise, for instance, in spatially distributed models, where the state at each time is itself a function:

$$u(t) = u(t, \cdot).$$

Therefore, the natural state space is no longer $\mathbb{R}^N$, but a Banach space X.

What is gained

The abstract equation can describe problems related to

- nonlinear PDEs;
- abstract differential equations;
- evolution problems.

What becomes more difficult

In infinite dimension:

- bounded sets need not be relatively compact;
- Some topological methods change drastically, or simply cease to exist.

The periodic problem studied in the thesis

Let X be a real separable reflexive Banach space, with dual X^*. We study

$$\begin{cases} \dfrac{d}{dt} S(t, u'(t)) = f(t, u(t)), & t \in (0, T), \\ u(0) = u(T), \qquad u'(0) = u'(T), \end{cases} \tag{P}$$

where

$$S : [0, T] \times X \to X^*, \qquad f : [0, T] \times X \to X^*.$$

The same framework applies directly in a Hilbert space H, with $S, f : [0, T] \times H \to H$.

Definition of solution

A solution of (P) is a function $u \in C^1([0, T]; X)$ such that

$$S(\cdot, u'(\cdot)) \in W^{1,1}([0, T]; X^*)$$

and the differential equation holds a.e. on $(0, T)$.

Basic notation and periodic spaces

- Throughout this presentation,

$$\langle x^*, x \rangle$$

denotes the duality pairing between X^* and X.

- For a Banach space Y, we write

$$C_T([0,T];Y) := \{u \in C([0,T];Y) : u(0) = u(T)\},$$

and

$$C_0([0,T];Y) := \{v \in C([0,T];Y) : v(0) = v(T) = 0\}.$$

Structural assumptions on S

We assume that $S : [0,T] \times X \to X^*$ satisfies:

(S0) **Continuity, normalization, and periodicity.** The map S is continuous and, for every $t \in [0,T]$ and $x \in X$,

$$S(t,x) = 0 \iff x = 0, \qquad S(0,x) = S(T,x).$$

(S1a) **Local uniform strict monotonicity.** For every $M > 0$, there exists a continuous non-decreasing function

$$\omega_M : [0,\infty) \to [0,\infty), \qquad \omega_M(0) = 0, \quad \omega_M(r) > 0 \text{ for } r > 0,$$

such that, for every $t \in [0,T]$ and $\|x\|_X, \|y\|_X \leq M$,

$$\langle S(t,x) - S(t,y), x - y \rangle \geq \omega_M\Big(\|x - y\|_X\Big).$$

(S1b) **Uniform coercivity.** There exists a continuous non-decreasing function

$$\alpha : [0,\infty) \to [0,\infty), \qquad \alpha(r) \to \infty \text{ as } r \to \infty,$$

such that, for every $t \in [0,T]$ and $x \in X$,

$$\langle S(t,x), x \rangle \geq \alpha\Big(\|x\|_X\Big) \|x\|_X.$$

(S2) **Boundedness on bounded sets.** There exists a continuous function $\beta : [0,T] \times [0,\infty) \to [0,\infty)$ such that, for every $t \in [0,T]$ and $x \in X$,

$$\|S(t,x)\|_{X^*} \leq \beta\Big(t, \|x\|_X\Big).$$

An example: the $p(t)$-Laplacian-type operator

Let H be a real Hilbert space and let $\mathcal{R} : H \to H^*$ be the Riesz isomorphism.

Assume that $p \in C([0,T])$ satisfies

$$1 < p_- \leq p(t) \leq p_+ < \infty, \qquad p(0) = p(T).$$

Variable-exponent operator

Define

$$S(t,0) := 0, \qquad S(t,x) := \|x\|_H^{p(t)-2} \mathcal{R}x, \quad x \neq 0.$$

Then

$$S : [0,T] \times H \to H^*$$

satisfies **(S0)–(S2)**.

In the Hilbert-space formulation, we may work directly with

$$S(t,x) = \|x\|_H^{p(t)-2} x, \qquad S : [0,T] \times H \to H.$$

Invertibility and properties of the inverse

Fix $t \in [0, T]$ and set

$$S_t := S(t, \cdot) : X \to X^*.$$

Under **(S0)–(S2)**, S_t is continuous, strictly monotone, and coercive. Hence, by the Minty–Browder theorem, S_t is bijective.

We denote its inverse by

$$\Gamma(t, \cdot) := S_t^{-1} : X^* \to X.$$

Properties of Γ

- $\Gamma : [0, T] \times X^* \to X$ is jointly continuous and strictly monotone;
- $\|\Gamma(t, y)\|_X \to \infty$ as $\|y\|_{X^*} \to \infty$, uniformly in t;
- if $g_n \to g$ in $C([0, T]; X^*)$, then

$$\Gamma(\cdot, g_n(\cdot)) \to \Gamma(\cdot, g(\cdot)) \quad \text{in } C([0, T]; X).$$

Nonlinear Green operator and generalized reduction

Compatibility and first integration

Let u solve the *non-homogeneous periodic problem*,

$$\begin{cases} \dfrac{d}{dt} S(t, u'(t)) = h(t), \\ u(0) = u(T), \qquad u'(0) = u'(T), \end{cases} \qquad h \in L^1([0,T]; X^*). \tag{NH}$$

Define

$$Qh := \int_0^T h(t)\, dt, \qquad H(h)(t) := \int_0^t h(s)\, ds.$$

Integrating from 0 to T gives the following **necessary condition**

$$\boxed{Qh = S(T, u'(T)) - S(0, u'(0)) = 0.}$$

Integrating the differential equation from 0 to s, for $s \in [0,T]$, then applying $\Gamma(s,\cdot)$ and finally integrating from 0 to t, yields

$$\boxed{u(t) = u(0) + \int_0^t \Gamma\big(s, c + H(h)(s)\big)\, ds, \qquad c \in X^*}$$

Position periodicity and the implicit equation

Evaluating the representation formula at $t = T$, we obtain

$$u(T) - u(0) = \int_0^T \Gamma\big(t, c + H(h)(t)\big)\, dt.$$

Therefore,

$$u(T) = u(0) \quad \Longleftrightarrow \quad \int_0^T \Gamma\big(t, c + H(h)(t)\big)\, dt = 0.$$

To express this condition, for each $g \in C([0,T]; X^*)$, define

$$\Phi_g : X^* \to X, \qquad \Phi_g(c) := \int_0^T \Gamma\big(t, c + g(t)\big)\, dt.$$

Hence, the periodicity condition is equivalent to requiring the integration constant $c \in X^*$ to solve

$$\boxed{\Phi_{H(h)}(c) = 0.}$$

Solvability of the implicit equation

Key lemma

For every $g \in C([0,T];X^*)$, there exists a unique $\widehat{c}(g) \in X^*$ such that

$$\Phi_g\big(\widehat{c}(g)\big) = 0.$$

Moreover, $\widehat{c} : C([0,T];X^*) \to X^*$ is continuous and maps bounded sets into bounded sets.

The nonlinear Green operator

From the previous construction, the integration constant is $c = \widehat{c}\big(H(h)\big)$. This leads us to define

$$\mathcal{T}(h)(t) := \int_0^t \Gamma\Big(s, \widehat{c}\big(H(h)\big) + H(h)(s)\Big)\, ds.$$

Since $\widehat{c}(H(h))$ solves $\Phi_{H(h)}(c) = 0$, we have

$$\mathcal{T}(h)(0) = \mathcal{T}(h)(T) = 0.$$

Therefore $\mathcal{T} : L^1([0,T]; X^*) \longrightarrow C_0([0,T]; X)$.

Solvability and representation

The problem (NH) has a solution if and only if there exists $u \in C_T([0,T]; X)$ such that

$$\begin{cases} u(t) = u(0) + \mathcal{T}(h)(t), & t \in [0,T] \\ Qh = 0. \end{cases}$$

From the main problem to an abstract system

We now return to the nonlinear periodic problem

$$\begin{cases} \dfrac{d}{dt} S(t, u'(t)) = f(t, u(t)), \\ u(0) = u(T), \qquad u'(0) = u'(T). \end{cases}$$

Assume that f is Carathéodory and satisfies the local growth conditions needed to define the Nemytskii operator $N_f : C([0,T]; X) \to L^1([0,T]; X^*)$, $N_f(u)(t) := f(t, u(t))$.

Applying the previous representation with $h = N_f(u)$, we obtain

$$\boxed{\begin{cases} u = u(0) + \mathcal{T}\big(N_f(u)\big), \\ QN_f(u) = 0, \end{cases}}$$

The compactness defect

In infinite dimension, $u \mapsto u(0)$ is not a completely continuous operator. Hence, the constant component $u(0)$ must be separated.

The generalized Lyapunov–Schmidt decomposition

For every $u \in C_T([0,T];X)$, we have the following unique decomposition,

$$u = u(0) + \big(u - u(0)\big), \qquad u(0) \in X, \quad u - u(0) \in C_0([0,T];X).$$

Since $X \cap C_0([0,T];X) = \{0\}$ (using X represented as the constant functions),

$$C_T([0,T];X) = X \oplus C_0([0,T];X).$$

Set

$$a := u(0), \qquad v := u - a.$$

Substituting $u = a + v$ into the abstract system and cancelling the constant component a, we obtain

$$\boxed{\begin{cases} v = \mathcal{T}\big(N_f(a+v)\big), \\ QN_f(a+v) = 0, \end{cases} \qquad a \in X, \quad v \in C_0([0,T];X).} \tag{LS}$$

Two existence strategies

The two methods developed in this thesis differ in how they treat the averaged constraint

$$QN_f(a+v) = 0,$$

which is a possibly infinite–dimensional problem.

First method: eliminate the constant component

If the constraint determines

$$a = \widehat{A}_f(v),$$

then the system reduces to a single compact fixed-point equation for v.

Second method: approximate the coupled system

When the constraint cannot be solved explicitly for a, we solve finite-dimensional approximations of the full system and then pass to the limit to recover a solution of the original problem.

First existence method: mean-admissible nonlinearities

Mean-admissibility and the averaged constraint

Let Y, Z be reflexive Banach spaces and $f : [0,T] \times Y \to Z$. For $v \in C_0([0,T];Y)$, define

$$F_v(a) := \int_0^T f\big(t, a + v(t)\big)\, dt.$$

Definition

The map f is *mean-admissible* if, for every v,

$$F_v(a) = 0$$

has a unique solution $a = \widehat{A}_f(v)$, where $\widehat{A}_f : C_0([0,T];Y) \to Y$ is continuous and maps bounded sets into bounded sets.

In the generalized Lyapunov–Schmidt system, $QN_f(a+v) = F_v(a)$. Hence,

$$\boxed{QN_f(a+v) = 0 \quad \Longleftrightarrow \quad a = \widehat{A}_f(v).}$$

A sufficient criterion for mean-admissibility

The implicit equation for Γ suggests imposing analogous monotonicity and coercivity assumptions on f.

Proposition

Let Y be a real separable reflexive Banach space and let $f : [0, T] \times Y \to Y^*$ be continuous. Assume, uniformly in $t \in [0, T]$, that:

(i) There exists a non-decreasing function $\eta : [0, \infty) \to [0, \infty)$, with $\eta(r) > 0$ for $r > 0$, such that

$$\langle f(t,x) - f(t,y), x - y\rangle \geq \eta\big(\|x - y\|_Y\big).$$

(ii) There exist $\Theta : [0, \infty) \to \mathbb{R}$ and a non-decreasing function $\gamma : [0, \infty) \to [0, \infty)$ such that

$$\langle f(t,x), x\rangle \geq \Theta(\|x\|_Y)\,\|x\|_Y\,, \qquad \|f(t,x)\|_{Y^*} \leq \gamma(\|x\|_Y),$$

and

$$\Theta(r) \to \infty, \qquad \Theta(r)r - M\gamma(r) \to \infty \quad \text{for every } M > 0.$$

Then f is *mean-admissible*.

The criterion is sufficient, but not necessary

Let $T = 2\pi$, let H be a real Hilbert space, and consider

$$f(t,x) = m(t)x + b(t), \qquad m(t) := 1 + 2\sin t.$$

Since $m(t) < 0$ for some t, the map $f(t,\cdot)$ is not pointwise monotone.

However, the averaged equation

$$\int_0^{2\pi} f\big(t, a + v(t)\big)\, dt = 0$$

has the unique solution

$$\widehat{A}_f(v) = -\frac{1}{2\pi}\left(\int_0^{2\pi} m(t)v(t)\, dt + \int_0^{2\pi} b(t)\, dt\right).$$

Fixed-point reduction and the compactness issue

If f is mean-admissible, then the averaged constraint satisfies

$$QN_f(a+v)=0 \quad \iff \quad a=\widehat{A}_f(v).$$

Therefore, the generalized Lyapunov–Schmidt system reduces to

$$\boxed{v=\mathcal{T}\Big(N_f\big(\widehat{A}_f(v)+v\big)\Big), \qquad v\in C_0([0,T];X).} \tag{FP$_X$}$$

To apply topological methods such as Leray–Schauder degree, the right-hand side must define a completely continuous operator.

However, for an arbitrary Banach space X,

$$\mathcal{T}:L^1([0,T];X^*)\to C_0([0,T];X)$$

need not be completely continuous.

Recovering compactness through a Gelfand triple

To recover compactness, we assume that X is part of a compact Gelfand triple

$$X \overset{i}{\hookrightarrow} H \cong H^* \overset{i^*}{\hookrightarrow} X^*,$$

where H is a real Hilbert space and $i : X \hookrightarrow H$ is dense and compact.

H-valued periodic solutions

Let $f : [0,T] \times H \to H$. An H-valued periodic solution is a function

$$u = a + v \in C_T([0,T];H), \qquad a \in H, \quad v \in C_0([0,T];X) \cap C^1([0,T];X),$$

satisfying

$$\begin{cases} \dfrac{d}{dt} S(t, u'(t)) = i^* f(t, u(t)), \\ u(0) = u(T), \qquad u'(0) = u'(T). \end{cases}$$

The compact fixed-point problem

The compact embedding $X \hookrightarrow H$ allows us to regard the Green operator as

$$\mathcal{T} : L^1([0,T]; X^*) \to C_0([0,T]; H).$$

Compactness lemma

The operator $\mathcal{T}$ is continuous and maps bounded sets into relatively compact subsets of $C_0([0,T]; H)$.

Consequently, the reduced equation in the Gelfand setting is

$$\boxed{v = \mathcal{T}\Big(i^* N_f\left(\widehat{A}_f(v) + v\right)\Big), \qquad v \in C_0([0,T]; H).} \tag{FP$_H$}$$

Its right-hand side is completely continuous, so Leray–Schauder degree can be applied.

A continuation theorem

Theorem

Let $\Omega \subset C_T([0,T];H)$ be bounded and open, and let $f : [0,T] \times H \to H$ be a *mean-admissible* Carathéodory function. Assume that:

1. for every $\lambda \in (0,1)$, the problem

$$\begin{cases} \dfrac{d}{dt} S(t, u'(t)) = \lambda i^* f(t, u(t)), \\ u(0) = u(T), \qquad u'(0) = u'(T), \end{cases}$$

   has no H-valued periodic solution on $\partial\Omega$.

2. there exists $c \in H$ such that the constant function $t \mapsto c$ belongs to Ω and

$$\int_0^T f(t,c)\,dt = 0;$$

Then (P) admits an H-valued periodic solution $u \in \overline{\Omega}$.

Proof idea I: constraint reduction

- We will start by supposing that there are no H-valued solutions in $\partial\Omega$ for $\lambda = 1$, since otherwise we would be finished with the proof.
- Define the constraint set
$$M := \left\{ u \in C_T([0,T];H) : \int_0^T f(t,u(t))\,dt = 0 \right\}.$$
which is nonempty since $u(t) \equiv c \in M$.
- Mean-admissibility gives a parametrization
$$R : C_0([0,T];H) \to M, \qquad R(v) := \widehat{A}_f(v) + v.$$
Since $v(0) = 0$, one has $R^{-1}(u) = u - u(0)$, so R is a homeomorphism.
- Transfer the domain through R:
$$\Omega_0 := R^{-1}(\Omega \cap M).$$
Then Ω_0 is bounded and open. Moreover,
$$\widehat{A}_f(0) = c \qquad \Longrightarrow \qquad 0 \in \Omega_0.$$

Proof idea II: compact homotopy and degree

- For $\lambda \in [0,1]$, define
$$\mathcal{G}(\lambda, v) := \mathcal{T}\Big(\lambda i^* N_f(R(v))\Big).$$
The compactness of $\mathcal{T}$ implies that $\mathcal{G}$ is completely continuous.

- If $v = \mathcal{G}(\lambda, v)$ with $v \in \partial\Omega_0$ and $\lambda \in (0,1]$, then $u := R(v)$ solves the λ-problem and belongs to $\partial\Omega$, contradicting the boundary assumption.

- At $\lambda = 0$, $\mathcal{G}(0,v) = \mathcal{T}(0) = 0.$ Since $0 \in \Omega_0$, there are no boundary fixed points, and homotopy invariance gives
$$d_{LS}\big[I - \mathcal{G}(1,\cdot), \Omega_0, 0\big] = d_{LS}\big[I, \Omega_0, 0\big] = 1.$$

- Since the degree is different from zero, there exists $v \in \Omega_0$ with $v = \mathcal{G}(1,v).$ Therefore
$$u := R(v) = \widehat{A}_f(v) + v$$
is an H-valued periodic solution with $u \in \Omega$.

Application: a double-phase periodic problem

Let $\mathcal{O} \subset \mathbb{R}^d$ be open, bounded with Lipschitz boundary, and set

$$X = W_0^{1,\beta}(\mathcal{O}) \hookrightarrow H = L^2(\mathcal{O}) \hookrightarrow X^* = W^{-1,\beta'}(\mathcal{O}), \qquad \beta > 2.$$

Let $p \in C([0, 2\pi])$ such that

$$1 < p_- \leq p(t) \leq p_+ \leq \beta, \qquad p(0) = p(2\pi),$$

and define

$$m(t) := 1 + 2\sin t, \qquad b \in C([0, 2\pi]; H).$$

Consider, in X^*,

$$\begin{cases} \partial_t \left[-\operatorname{div}\left(|\nabla u_t|^{p(t)-2} \nabla u_t + |\nabla u_t|^{\beta-2} \nabla u_t \right) \right] = i^* \left(m(t) u + b(t) \right), \\ u|_{\partial\mathcal{O}} = 0, \qquad u(0) = u(2\pi), \qquad u_t(0) = u_t(2\pi). \end{cases}$$

All the assumptions of the continuation theorem can be verified. Therefore, the problem admits an H-valued periodic solution.

Second existence method: Hartman–Galerkin

From monotonicity to the Galerkin strategy

The monotone structure of S suggests approaching the problem through monotone-operator methods.

In infinite dimension, a natural strategy is to approximate each problem by finite-dimensional problems, solve the corresponding projected problems and verify that this sequence converges to a solution of the original problem. This is the Galerkin method.

$$\boxed{\text{finite-dimensional approximation} \longrightarrow \text{solutions } u_N \longrightarrow \text{uniform estimates} \longrightarrow \text{limit solution}}$$

Variable-exponent functional setting

Let $p \in C([0,T])$ satisfy

$$1 < p_- \leq p(t) \leq p_+ < \infty, \qquad p(0) = p(T),$$

and define

$$p'(t) := \frac{p(t)}{p(t)-1}.$$

We work in the reflexive space

$$V := W_T^{1,p(\cdot)}([0,T];X) = \left\{ u \in W^{1,p(\cdot)}([0,T];X) : u(0) = u(T) \right\}.$$

For this second existence method, $S : [0,T] \times X \to X^*$ is assumed to satisfy the following stronger hypotheses.

(H0) Continuity, normalization, and periodicity. The map S is continuous and satisfies

$$S(t,x) = 0 \iff x = 0, \qquad S(0,x) = S(T,x),$$

for every $t \in [0,T]$ and $x \in X$.

Stronger hypotheses on the principal operator

(H1) Local uniform strict monotonicity. For every $M > 0$, there exists a continuous non-decreasing function

$$\omega_M : [0, \infty) \to [0, \infty), \qquad \omega_M(0) = 0, \quad \omega_M(r) > 0 \text{ for } r > 0,$$

such that

$$\langle S(t,x) - S(t,y), x - y \rangle \geq \omega_M\left(\|x - y\|_X \right)$$

for every $t \in [0,T]$ and $\|x\|_X, \|y\|_X \leq M$.

(H2) Variable-exponent coercivity. There exists $c_0 > 0$ such that

$$\langle S(t,x), x \rangle \geq c_0 \|x\|_X^{p(t)}$$

for every $t \in [0,T]$ and $x \in X$.

(H3) Variable-exponent growth. There exists $c_1 > 0$ such that

$$\|S(t,x)\|_{X^*} \leq c_1 \left(1 + \|x\|_X^{p(t)-1} \right)$$

for every $t \in [0,T]$ and $x \in X$.

Assumptions on the nonlinearity

Let $m, k : [0,T] \times X \to X^*$ be Carathéodory maps and set

$$f(t,x) := m(t,x) + k(t,x).$$

(F1) Local growth. For every $r > 0$, there exists $\eta_r \in L^{p'(\cdot)}([0,T])$ such that, for a.e. $t \in [0,T]$,

$$\|m(t,x)\|_{X^*} + \|k(t,x)\|_{X^*} \leq \eta_r(t) \qquad \text{whenever } \|x\|_X \leq r.$$

(F2$_m$) Monotonicity. For a.e. t,

$$\langle m(t,x) - m(t,y), x - y \rangle \geq 0.$$

(F2$_{ws}$) Weak-to-strong continuity. For a.e. t,

$$x_n \rightharpoonup x \quad \Longrightarrow \quad k(t,x_n) \to k(t,x) \quad \text{in } X^*.$$

(F3$_e$) Hartman condition. There exists $R > 0$ such that, for a.e. $t \in [0,T]$,

$$\langle f(t,x), x \rangle \geq 0 \qquad \text{whenever } \|x\|_X \geq R.$$

Main Hartman-type theorem

Theorem

Assume that S satisfies **(H0)–(H3)**, and that

$$f(t,x) = m(t,x) + k(t,x)$$

satisfies **(F1)**, **(F2$_m$)**, **(F2$_{ws}$)**, and **(F3$_e$)**.

Then (P) admits at least one periodic solution $u \in C^1([0,T];X)$.

The finite-dimensional framework

Since X is separable, choose a linearly independent sequence

$$(e_j)_{j\geq 1} \subset X, \qquad \overline{\operatorname{span}\{e_j : j \geq 1\}}^{X} = X,$$

and set

$$X_N := \operatorname{span}\{e_1, \dots, e_N\}.$$

Endow X_N with the auxiliary inner product $(\cdot,\cdot)_N$ for which $(e_1, \dots, e_N)$ is orthonormal, and define

$$R_N y := \sum_{j=1}^{N} \langle y, e_j \rangle \, e_j, \quad y \in X^*,$$

$$S_N(t,x) := R_N S(t,x), \quad x \in X_N.$$

Projected S operator

For every N and t, $S_N(t,\cdot) : X_N \to X_N$ is a homeomorphism. Moreover, $\Gamma_N(t,\cdot) := (S_N(t,\cdot))^{-1}$ verifies all the properties of Γ.

The regularized projected problems

To obtain a strict exterior inequality while recovering the original problem as $N \to \infty$, set

$$\varepsilon_N := \frac{1}{N}, \qquad \widetilde{f}_N(t,x) := f(t,x) + \varepsilon_N S(t,x).$$

We want to find solutions u_N for the problem

$$\begin{cases} \dfrac{d}{dt} S_N(t, u_N'(t)) = R_N \widetilde{f}_N(t, u_N(t)), \\ u_N(0) = u_N(T), \qquad u_N'(0) = u_N'(T). \end{cases} \tag{P_N}$$

Hence, we will consider, for $\lambda \in [0,1]$, the one–parameter family of problems,

$$\begin{cases} \dfrac{d}{dt} S_N(t, u_N'(t)) = \lambda R_N \widetilde{f}_N(t, u_N(t)), \\ u_N(0) = u_N(T), \qquad u_N'(0) = u_N'(T). \end{cases} \tag{$\mathrm{P}_{N,\lambda}$}$$

Finding the projected solutions by degree theory

Write

$$u_N = a_N + v_N, \quad a_N = u_N(0), \quad v_N = u_N - a_N,$$

hence $a_N \in X_N$, $v_N \in E_N := C_0([0,T];X_N)$, where $\|v\|_{E_N} := \max_{t\in[0,T]} \|v(t)\|_X$.

The projected perturbated problem $(\mathrm{P}_{N,\lambda})$ is equivalent to

$$\begin{cases} v_N = \mathcal{T}_N\Big(\lambda R_N N_{\widetilde{f}_N}(a_N + v_N)\Big), \\ \mathcal{Q}_N(a_N + v_N) = 0, \end{cases}$$

where

$$\mathcal{Q}_N(u) := R_N \int_0^T \widetilde{f}_N(t, u(t))\, dt.$$

Define the associated fixed-point operator $\mathcal{K}_{N,\lambda} : X_N \times E_N \to X_N \times E_N$ as:

$$\mathcal{K}_{N,\lambda}(a,v) := \Big(a - \mathcal{Q}_N(a+v),\, \mathcal{T}_N\Big(\lambda R_N N_{\widetilde{f}_N}(a+v)\Big)\Big).$$

This operator is completely continuous since X_N is finite-dimensional (standard argument using the properties of Γ_N and $\mathcal{Q}_N$).

Proof idea I: uniform Hartman bounds

- Let u_N solve $(P_{N,\lambda})$. Testing the equation with u_N gives

$$c_0 \int_0^T \left\| u_N' \right\|_X^{p(t)} dt \leq -\lambda \int_0^T \left\langle \widetilde{f}_N(t, u_N), u_N \right\rangle dt.$$

- The Hartman condition and the strict regularization imply

$$\exists\, \tau_N \in [0,T] \qquad \|u_N(\tau_N)\|_X < R.$$

  Moreover, the integrand is nonpositive when $\|u_N(t)\|_X \geq R$, while **(F1)** controls it inside the ball, hence giving a uniform bound that only depends on R and **(F1)**.

- Consequently $\|u_N'\|_{L^{p(\cdot)}([0,T];X)} \leq C$, and by the FTC,

$$\|u_N(t)\|_X \leq \|u_N(\tau_N)\|_X + \int_{\tau_N}^t \|u_N'(s)\|_X \, ds.$$

  Using $\|u_N(\tau_N)\|_X < R$ and Hölder's inequality, we obtain, for a constant independent of N and λ,

$$\|u_N\|_{C([0,T];X)} \leq C \qquad \Longrightarrow \qquad \|u_N\|_V \leq C'.$$

Proof idea II: degree and projected solutions

The uniform estimate for u_N over $C([0,T];X)$ gives

$$\|a_N\|_X + \|v_N\|_{E_N} \leq C_1.$$

Choose $\rho > \max\{C_1, R\}$, and define the following open bounded set

$$\Omega_N := B_\rho^{X_N} \times B_\rho^{E_N} \subset X_N \times E_N,$$

where $B_\rho^{X_N} := \{a \in X_N : \|a\|_X < \rho\}$.

- No fixed point of $\mathcal{K}_{N,\lambda}$ lies on $\partial\Omega_N$ for $\lambda \in (0,1]$.
- At $\lambda = 0$ the fixed point problem is $(a,v) = (a - \mathcal{Q}_N(a+v), 0)$; hence, the problem is equivalent to $\mathcal{Q}_N(a) = 0$. By an auxiliary homotopy and the exterior inequality, we get

$$d_{LS}[I - \mathcal{K}_{N,1}, \Omega_N, 0] = d_{LS}[I - \mathcal{K}_{N,0}, \Omega_N, 0] = d_B[\mathcal{Q}_N, B_\rho^{X_N}, 0] = 1.$$

- Hence $\mathcal{K}_{N,1}$ has a fixed point, and the regularized projected problem admits a solution u_N.

Proof idea III: passage to the limit

- Since $\|u_N\|_V \leq C$ uniformly, and V is reflexive, up to a subsequence,

$$u_N \rightharpoonup u \quad \text{in } V, \qquad u_N(t) \rightharpoonup u(t) \quad \text{in } X.$$

- Weak-to-strong continuity and local growth give

$$k(\cdot, u_N) \to k(\cdot, u) \quad \text{in } L^{p'(\cdot)}([0,T]; X^*), \qquad \varepsilon_N S(\cdot, u_N) \to 0.$$

- Using the monotonicity of S and m, Minty's trick identifies the weak limit and yields

$$\int_0^T \left\langle S(t, u'), \varphi' \right\rangle \, dt + \int_0^T \langle f(t,u), \varphi \rangle \, dt = 0, \qquad \forall \varphi \in V.$$

- Thus, u is a weak periodic solution. Setting $q := S(\cdot, u')$, we have

$$q \in W^{1,p'(\cdot)}([0,T]; X^*), \qquad u'(t) = \Gamma(t, q(t)) \in C([0,T]; X),$$

and therefore $u \in C^1([0,T]; X)$. The periodicity of u follows from the same equation.

Two limiting cases

Purely monotone nonlinearity

Assume that S satisfies **(H0)–(H3)**, and that $f : [0,T] \times X \to X^*$ satisfies **(F1)** and is monotone in its second variable. If, for some $R > 0$,

$$\langle f(t,x), x\rangle \geq 0 \qquad \text{for a.e. } t \in [0,T] \text{ and } \|x\|_X = R,$$

then the periodic problem admits a solution. Any two solutions differ by a constant element of X; if $f(t,\cdot)$ is strictly monotone, the solution is unique.

Purely weak-to-strong nonlinearity

Suppose that X is infinite-dimensional. If f satisfies **(F1)**, is sequentially weak-to-strong continuous, and

$$\langle f(t,x), x\rangle \geq 0 \qquad \text{for } \|x\|_X \geq R,$$

then every periodic solution is stationary. In particular, $u \equiv 0$ is a solution.

Radial Hilbert-space operators

Let X be a Hilbert space and assume that S satisfies **(H0)–(H3)** and has the radial form

$$S(t,0) = 0, \qquad S(t,x) = \beta(t, \|x\|_X)\mathcal{R}x, \qquad \beta(t,r) > 0.$$

Proposition

Under the hypotheses of the main Hartman-type theorem, the periodic problem admits a solution u satisfying

$$\|u(t)\|_X \leq R, \qquad t \in [0,T].$$

This partially extends the corresponding finite-dimensional radial Hartman bound of [8] to the Hilbert-space setting.

Application: a nonlocal Sobolev-type problem

Let $\mathcal{O} \subset \mathbb{R}^d$ be defined as the previous application, $X = H_0^1(\mathcal{O})$, $X^* = H^{-1}(\mathcal{O})$, $A = -\Delta$ (in weak form), and

$$S(t,x) = \|x\|_X^{p(t)-2} Ax.$$

Let $b \in C([0,T]; H^{-1}(\mathcal{O}))$ be nonconstant and set

$$B := \|b\|_{C([0,T];H^{-1}(\mathcal{O}))} < 1.$$

We study

$$\begin{cases} \partial_t \left[- \|\nabla u_t(t,\cdot)\|_2^{p(t)-2} \Delta_x u_t(t,x) \right] = - \dfrac{\Delta_x u(t,x)}{\sqrt{1 + \|\nabla u(t,\cdot)\|_2^2}} + b(t), \\ u|_{\partial\mathcal{O}}(t,\cdot) = 0, \qquad u(0) = u(T), \quad u_t(0) = u_t(T). \end{cases}$$

Then there exists a unique nonstationary periodic solution satisfying

$$\|\nabla u(t,\cdot)\|_{L^2(\mathcal{O})} \le \frac{B}{\sqrt{1-B^2}}, \qquad t \in [0,T].$$

Concluding remarks

Conclusions

- We introduced a broad class of monotone operators S, including p-Laplacian, $p(t)$-Laplacian, and double-phase type operators.
- We derived a nonlinear Green operator for the associated nonhomogeneous periodic problem,
- For the nonlinear periodic problem, we found an equivalent formulation in the form of a generalized Lyapunov–Schmidt system.
- Two existence mechanisms were developed:

$\boxed{\text{mean-admissibility}}$ $\boxed{\text{Hartman–Galerkin}}$

- The results cover many types of problems, involving different nonlinearities and differential operators.

Open problems and future directions

Derivative-dependent nonlinearities

Can the existence methods developed here be extended to

$$\frac{d}{dt}S(t, u'(t)) = f(t, u(t), u'(t))?$$

This would require new compactness arguments, a priori estimates for u', and possibly additional structural conditions on f.

Alternative criteria for mean-admissibility

Are there other sufficient conditions for mean-admissibility beyond the pointwise monotonicity and coercivity assumptions considered here?

Further applications of the decomposition

Can the decomposition underlying the generalized Lyapunov–Schmidt system be adapted to other problems with a noncompact or infinite-dimensional constant component?

Bibliography I

C. D. Aliprantis and K. C. Border, *Infinite Dimensional Analysis: A Hitchhiker's Guide*, 3rd ed., Springer, Berlin–Heidelberg, 2006.

V. Barbu, *Nonlinear Differential Equations of Monotone Types in Banach Spaces*, Springer Monographs in Mathematics, Springer, New York, 2010.

H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, Universitext, Springer, New York, 2011.

K. Deimling, *Nonlinear Functional Analysis*, Springer-Verlag, Berlin, 1985.

L. Diening, P. Harjulehto, P. Hästö and M. Růžička, *Lebesgue and Sobolev Spaces with Variable Exponents*, Lecture Notes in Mathematics, vol. 2017, Springer, Berlin–Heidelberg, 2011.

G. Dinca and J. Mawhin, *Brouwer Degree: The Core of Nonlinear Analysis*, Progress in Nonlinear Differential Equations and Their Applications, vol. 95, Birkhäuser, Cham, 2021.

Bibliography II

- X.-L. Fan and X. Fan, A Knobloch-type result for $p(t)$-Laplacian systems, *J. Math. Anal. Appl.* 282 (2003), 453–464.

- M. García-Huidobro, R. Manásevich, J. Mawhin and S. Tanaka, Periodic solutions for nonlinear systems of ODEs with generalized variable exponents operators, *J. Differential Equations* 388 (2024), 34–58.

- M. García-Huidobro, R. Manásevich, J. Mawhin and S. Tanaka, Two point boundary value problems for ordinary differential systems with generalized variable exponents operators, *Nonlinear Anal. Real World Appl.* 81 (2025), 104196.

- M. A. Krasnosel'skii and Ya. B. Rutickii, *Convex Functions and Orlicz Spaces*, Noordhoff, Groningen, 1961.

- R. Manásevich and J. Mawhin, Periodic solutions for nonlinear systems with p-Laplacian-like operators, *J. Differential Equations* 145 (1998), 367–393.

Bibliography III

R. Manásevich and J. Mawhin, Boundary value problems for nonlinear perturbations of vector p-Laplacian-like operators, *J. Korean Math. Soc.* 37 (2000), 665–685.

R. Manásevich and J. Novoa, Global continuum of solutions for systems of ODEs with periodic boundary conditions and generalized variable-exponent operators, *Commun. Pure Appl. Anal.* 30 (2026), 151–166.

J. Mawhin, Two point boundary value problems for nonlinear second order differential equations in Hilbert spaces, *Tohoku Math. J.* 32 (1980), 225–233.

J. Mawhin, Leray–Schauder continuation theorems in the absence of a priori bounds, *Topol. Methods Nonlinear Anal.* 9 (1997), 179–200.

J. Mawhin and M. Willem, Periodic solutions of nonlinear differential equations in Hilbert spaces, in *Equadiff 78*, Università di Firenze, Firenze, 1978, pp. 323–332.

Bibliography IV

- J. Mawhin and M. Willem, Periodic solutions of some nonlinear second order differential equations in Hilbert spaces, in R. Conti (ed.), *Recent Advances in Differential Equations*, Academic Press, New York, 1981, pp. 287–294.
- C. H. Morales, A Bolzano's theorem in the new millennium, *Nonlinear Anal.* 51 (2002), 679–691.
- P. H. Rabinowitz, *Théorie du degré topologique et applications à des problèmes aux limites non linéaires*, Laboratoire d'Analyse Numérique, Université Paris VI, Paris, 1975. Lecture notes written by H. Berestycki.
- M. M. Rao and Z. D. Ren, *Theory of Orlicz Spaces*, Pure and Applied Mathematics, vol. 146, Marcel Dekker, New York, 1991.
- V. Rădulescu and D. Repovš, *Partial Differential Equations with Variable Exponents*, CRC Press, Boca Raton, 2015.
- G. Teschl, *Nonlinear Functional Analysis*, University of Vienna, 2005.

Bibliography V

- R. Vandervorst, *Topological Methods for Differential Equations: Degree Theory, Conley Index and Morse Theory*, lecture notes, VU University, 2014.
- Q. Zhang, Existence of solutions for weighted $p(r)$-Laplacian system boundary value problems, *J. Math. Anal. Appl.* 327 (2007), 127–141.
- Q. Zhang, X. Liu and Z. Qiu, Existence of solutions and multiple solutions for a class of weighted $p(r)$-Laplacian system, *J. Math. Anal. Appl.* 355 (2009), 620–633.
- Q. Zhang, Z. Qiu and X. Liu, Existence of multiple solutions for weighted $p(r)$-Laplacian equation Dirichlet problems, *Nonlinear Anal.* 70 (2009), 3721–3729.
- Q. Zhang, Z. Qiu and X. Liu, Existence of solutions for prescribed variable exponent mean curvature system boundary value problem, *Nonlinear Anal.* 71 (2009), 2964–2975.

Thank you

Questions